

The full character space is countable

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.countable_characterSpace_of_nonUnital_singleton`

theorem

Uses separated unit eigenvectors to count the non-scalar characters, then restores the one scalar character.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed. Let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ be nonzero irreducible and represent the unique unitary-equivalence class of nonzero irreducible comparisons. If D is a norm-closed unital $*$ -subalgebra of $\tilde{A} = \mathbb{C} \oplus A$, then the entire character space X of D is countable. Here $\chi_\infty(\lambda, a) = \lambda$ denotes the scalar-coordinate character of $\tilde{A}$.

Assumptions

Commutativity of D is not an additional hypothesis in this theorem. Separability is imposed on H , not on A or D .

Conclusion

Countability includes the scalar character $\chi_\infty|_D$, as well as all the other characters.

Proof route

Choose a unit eigenvector for each non-scalar character, prove their pairwise orthogonality, count disjoint small balls using separability, and adjoin the scalar character.

▼ Proof steps

1. Let $Y = X \setminus \{\chi_\infty|_D\}$. For each $\chi \in Y$, apply A joint unit eigenvector for each non-scalar character with the same A, H, π, D and that χ . Closedness of D and its inequality from the scalar character supply the remaining hypotheses. Fix a unit vector η_χ for each character so that

$$\widehat{\pi}(d)\eta_\chi = \chi(d)\eta_\chi \quad (d \in D),$$

where $\widehat{\pi}(\lambda, a) = \lambda I + \pi(a)$.

2. If $\chi \neq \psi$, choose $d \in D$ with $\chi(d) \neq \psi(d)$. Put $T = \widehat{\pi}(d)$. Since D is $*$ -closed, the eigenvector equation also gives $T^*\eta_\psi = \overline{\psi(d)}\eta_\psi$. With the inner product linear in its second argument,

$$\overline{\psi(d)}\langle \eta_\chi, \eta_\psi \rangle = \langle \eta_\chi, T^*\eta_\psi \rangle = \langle T\eta_\chi, \eta_\psi \rangle = \overline{\chi(d)}\langle \eta_\chi, \eta_\psi \rangle.$$

The two coefficients differ, so $\langle \eta_\chi, \eta_\psi \rangle = 0$.

3. Consequently

$$\|\eta_\chi - \eta_\psi\|^2 = \|\eta_\chi\|^2 + \|\eta_\psi\|^2 = 2.$$

The open balls $B(\eta_\chi, 1/3)$ are therefore nonempty and pairwise disjoint. Choose a countable dense set in H ; each ball contains a point of it, and disjoint balls require different points. Thus Y is countable. This is the separated-ball argument in Countability of the non-scalar character space.

4. Finally

$$X = Y \cup \{\chi_\infty|_D\}.$$

Adding this one character to the countable set Y proves countability of the entire character space.

Main citations

- Exact theorem and proof
- A joint unit eigenvector for each non-scalar character
- Countability of the non-scalar character space
- The scalar-coordinate character

Lean source signature (exact)

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theorem countable_characterSpace_of_nonUnital_singleton [Nontrivial A]
  [TopologicalSpace.SeparableSpace H]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)
  (D : StarSubalgebra ℂ (Unitization ℂ A))
  [IsClosed (D : Set (Unitization ℂ A))] :
  Countable (WeakDual.characterSpace ℂ D)
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[TopologicalSpace.SeparableSpace H]	The representation space has a countable dense subset, used to count the disjoint balls.
(pi : NonUnitalCStarRepresentation A H)	The specified $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$; no unit-preservation equation is required.
(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
(D : StarSubalgebra ℂ (Unitization ℂ A))	The unital $*$ -subalgebra of $\tilde{A}$ whose characters are being counted; no commutativity binder occurs.

In the source	Mathematical meaning
<code>[IsClosed (D : Set (Unitization $\mathbb{C}$ A))]</code>	That D is norm closed.
<code>Countable (WeakDual.characterSpace $\mathbb{C}$ D)</code>	The set of all nonzero continuous complex characters of D is countable, including the scalar-coordinate character restricted to D .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration:

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